

Jaemoo Choi

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RESEARCH INTEREST

- **Generative AI**, Diffusion Models, Video/World Models, LLM fine-tuning, Distillation
- **AI for science**, Boltzmann distribution sampling
- **Optimal Control**, Optimal Transport, Schrödinger Bridge, Optimization

EDUCATION

Georgia Institute of Technology
Postdoctoral Associate
Advisor: Yongxin Chen

Atlanta, GA
09/2024 – present

Seoul National University (SNU)
Ph.D. *in Mathematical Sciences*
Advisor: Myungjoo Kang

Seoul, Korea
03/2018 – 08/2024

Seoul National University (SNU)
B.S. *in Mathematics Education*

Seoul, Korea
03/2014 – 02/2018

SELECTED PUBLICATIONS

- [1] **Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On the Importance of Likelihood Estimation Beyond Loss Design**
ICML, 2026
In reward-based fine-tuning for diffusion models, accurate likelihood estimation is more important than the precise choice of RL loss. ELBO-based estimators make fine-tuning substantially efficient and stable.
- [2] **QUATRO: Query-Adaptive Trust Region Policy Optimization for LLM Fine-tuning**
ICLR, 2026
This work gives a theoretically grounded view of GRPO-style LLM fine-tuning by deriving an exact query-wise trust-region bound objective. This leads to more stable updates and helps reduce mode collapse.
- [3] **Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization**
ICML, 2026 [Spotlight, 2.2%]
We combine proximal optimization, cross-entropy objective, and ELBO-based likelihood estimation together for fine-tuning discrete diffusion LLMs.
- [4] **Adjoint Schrödinger Bridge Sampler**
NeurIPS, 2025 [Oral, 0.3%]
We introduce an efficient first-order method for reward-based diffusion sampler by extending adjoint matching. The method improves exploration and mode coverage in difficult sampling problems.
- [5] **Discrete Adjoint Schrödinger Bridge Sampler**
ICML, 2026
We develop a discrete-state counterpart of first-order reward-based diffusion sampler methods such as adjoint matching and adjoint Schrödinger bridge sampler. This makes principled sampling possible even in discrete state spaces where standard gradients do not exist.

INDUSTRY COLLABORATION PROJECTS

Scalable Reinforcement Learning for Diffusion Models NVIDIA, 2026

- We developed a simple trajectory-free method that incorporates reward feedback into velocity-matching.
- Developed an pipeline for standard video diffusion fine-tuning.

Diffusion Samplers with Force Fields and Energy Functions FAIR, Meta, 2025

- Developed a stable first-order framework for diffusion-based sampling using Schrödinger bridge methods.
- Studied sampling under force fields and energy-based objectives, with a focus on efficiency and stability.

Automatic Feature Extraction of Sensor Data Samsung Electronics, 2020–2021

- Developed machine learning methods for automatic feature extraction from sensor data.
- Improved downstream analysis and decision-making for industrial sensing applications.

Automatic Real-World Video Enhancement four-by-four, 2022–2023

- Served as Project Manager for a project on learning-based denoising and super-resolution for real-world video enhancement, targeting 4K and 8K video content from K-dramas and films.
- Addressed practical degradation sources by modeling diverse noise patterns and developing patch-based, multi-frame, and stitching-based restoration pipelines.

HONORS & AWARDS

Gold Reviewer for ICML 2026	2026
InnoCORE Postdoctoral Fellowship	2025
Best Reviewer for NeurIPS 2024	2024
Nurturing Next-generation Researchers Postdoctoral Fellowship (\$ 45000)	2024
Outstanding TA Awards, SNU	2019
Brain Korea Research Scholarship	2018 – 2022, 2024
Beakwoon Academic Excellence Scholarship	2018
Academic Excellence Scholarship, SNU	2015 – 2017
Chungkwan Academic Excellence Scholarship, SNU	2014

PUBLICATIONS

*Equal contribution *Core contributors †Equal advising

Conference Papers

- [C1] Efficient Adjoint Matching for Fine-tuning Diffusion Models
[Spotlight]
 Jeongwoo Shin*, Dongsoo Shin*, Yuchen Zhu, Wei Guo, Yongxin Chen, Joonseok Lee†, Jaewoong Choi†, **Jaemoo Choi**†
 Advances in Neural Information Processing Systems (NeurIPS), 2026
- [C2] Coarse-to-Fine Compositional Diffusion for Long-Horizon Planning
 Byoungwoo Park, Utkarsh A. Mishra, **Jaemoo Choi**, Juho Lee, Yongxin Chen
 Advances in Neural Information Processing Systems (NeurIPS), 2026
- [C3] Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On the Importance of Likelihood Estimation Beyond Loss Design
Jaemoo Choi*, Yuchen Zhu*, Wei Guo, Petr Molodyk, Bo Yuan, Jinbin Bai, Yi Xin, Molei Tao, Yongxin Chen
 International Conference on Machine Learning (ICML), 2026

- [C4] QUATRO: Query-Adaptive Trust Region Policy Optimization for LLM Fine-tuning
Doyeon Lee, Eunyi Lyou, Hyunsoo Cho[†], Sookyoung Kim[†], Joonseok Lee[†], **Jaemoon Choi**[†]
International Conference on Machine Learning (ICML), 2026
- [C5] Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization
[Spotlight, 2.0%]
Yuchen Zhu*, Wei Guo*, **Jaemoon Choi**, Petr Molodyk, Bo Yuan, Molei Tao, Yongxin Chen
International Conference on Machine Learning (ICML), 2026
- [C6] Discrete Adjoint Schrödinger Bridge Sampler
Wei Guo*, Yuchen Zhu, Xiaochen Du, Juno Nam, Yongxin Chen[†], Rafael Gómez-Bombarelli[†],
Guan-Horng Liu[†], Molei Tao[†], **Jaemoon Choi***
International Conference on Machine Learning (ICML), 2026
- [C7] Generalized Schrödinger Bridge on Graphs
Panagiotis Theodoropoulos, Juno Nam, Evangelos Theodorou[†], **Jaemoon Choi**[†]
International Conference on Machine Learning (ICML), 2026
- [C8] Efficient Generative Modeling beyond Memoryless Diffusion via Adjoint Schrödinger Bridge Matching
Jeongwoo Shin, Jinhwan Sul, Jaewoong Choi[†], Joonseok Lee[†], **Jaemoon Choi**[†]
International Conference on Machine Learning (ICML), 2026
- [C9] MetaDNS: Enhancing Exploration in Discrete Neural Samplers via Metadynamics
Xiaochen Du, Juno Nam, **Jaemoon Choi**, Wei Guo, Sathya Edamadaka, Junyi Sha, Elton Pan,
Yongxin Chen, Molei Tao, Rafael Gomez-Bombarelli
International Conference on Machine Learning (ICML), 2026
- [C10] Proximal Diffusion Neural Samplers
Wei Guo*, **Jaemoon Choi***, Yuchen Zhu, Molei Tao, Yongxin Chen
International Conference on Learning Representations (ICLR), 2026
- [C11] Adjoint Schrödinger Bridge Sampler [Oral, 0.3%]
Guan-Horng Liu*, **Jaemoon Choi***, Yongxin Chen, Benjamin Kurt Miller, Ricky T. Q. Chen*
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [C12] Non-equilibrium Annealed Adjoint Sampler
Jaemoon Choi*, Yongxin Chen, Molei Tao, Guan-Horng Liu*
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [C13] MDNS: Masked Diffusion Neural Sampler via Stochastic Optimal Control
Yuchen Zhu*, Wei Guo*, **Jaemoon Choi**, Guan-Horng Liu, Yongxin Chen[†], Molei Tao[†]
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [C14] Overcoming Spurious Solutions in Semi-Dual Neural Optimal Transport: A Smoothing Approach
for Learning the Optimal Transport Plan
Jaemoon Choi, Jaewoong Choi[†], Dohyun Kwon[†]
International Conference on Machine Learning (ICML), 2025
- [C15] Unsupervised Point Cloud Completion through Unbalanced Optimal Transport
Taekyung Lee, **Jaemoon Choi**, Myungjoo Kang[†], Jaewoong Choi[†]
International Conference on Machine Learning (ICML), 2025
- [C16] Robust Barycenter Estimation using Semi-Unbalanced Neural Optimal Transport
Milena Gazdieva*, **Jaemoon Choi***, Alexander Kolesov, Jaewoong Choi, Petr Mokrov, Alexander
Korotin
International Conference on Learning Representations (ICLR), 2025
- [C17] Improving Neural Optimal Transport via Displacement Interpolation
Jaemoon Choi, Yongxin Chen, Jaewoong Choi
International Conference on Learning Representations (ICLR), 2025

- [C18] Scalable Wasserstein Gradient Flow for Generative Modeling through Unbalanced Optimal Transport
Jaemoo Choi*, Jaewoong Choi* and Myungjoo Kang
 International Conference on Machine Learning (ICML), 2024
- [C19] Analyzing and Improving OT-based Adversarial Networks
Jaemoo Choi*, Jaewoong Choi* and Myungjoo Kang
 International Conference on Learning Representations (ICLR), 2024
- [C20] Generative Modeling through the Semi-dual Formulation of Unbalanced Optimal Transport
Jaemoo Choi*, Jaewoong Choi* and Myungjoo Kang
 Advances in Neural Information Processing Systems (NeurIPS), 2023
- [C21] Restoration based Generative Models
Jaemoo Choi*, Yesom Park* and Myungjoo Kang
 International Conference on Machine Learning (ICML), 2023

Preprints

- [P1] Scaling Reinforcement Learning for Diffusion Models via Velocity Matching
Jaemoo choi, Wei Guo, Yuchen Zhu, Arash Vahdat, Molei Tao, Julius Berner, Yongxin Chen
 preprint, 2026
- [P2] Hard-Constrained Sampling on Embedded Riemannian Manifolds via Adjoint Schrödinger Bridges
 Mattia Mosso, **Jaemoo Choi**, Heng Yang
 preprint, 2026
- [P3] RMA: an Agentic System for Research-Level Mathematical Problems
 Zelin Zhao, Bo Yuan, **Jaemoo Choi**, Yongxin Chen
 preprint, 2026
- [P4] MFM-point: Multi-scale Flow Matching for Point Cloud Generation
 Petr Molodyk*, **Jaemoo Choi***, David W. Romero, Ming-Yu Liu, Yongxin Chen
 preprint, 2025

INVITED TALKS

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|---|---------|
| [T1] KAIST Kim Jaechul Graduate School,
Distributional Transport for Generalizable and Adaptive AI | 08/2026 |
| [T2] CAU-AI Core Technology Seminar, Chungang University,
Optimal Transport and Schrodinger Bridge Theory and it's Application | 08/2026 |
| [T3] Post-ICML Seminar, UNIST,
Optimal Transport and Schrodinger Bridge Theory and it's Application | 07/2026 |
| [T4] Guest Lecture for Special Topics, Georgia Tech
RL-based Fine-tuning Diffusion Models for Alignment | 03/2026 |
| [T5] Center for AI Natural Sciences, KIAS
Efficient and Scalable Samplers with Optimal Transport Perspective | 02/2025 |
| [T6] Optimal Transport Guest Lecture, Georgia Tech
Unbalanced Optimal Transport and its Application to Estimate Wasserstein Gradient Flow and the Robust Barycenter Distribution | 03/2025 |
| [T7] Level Set Seminar, UCLA
Algorithms for Optimal Transport, the Barycenter Problem, and Their Applications | 11/2024 |

[T8]	FLAIR Seminar, Georgia Tech Recent Works on Inverse Problem via Diffusion Models	10/2024
[T9]	Stanford Research Institute (SRI) Various Formulations of Optimal Transport Problems and Its Application to Generative Modeling	07/2024
[T10]	Center for AI Natural Sciences, KIAS Various Formulations of Optimal Transport Problems and Its Application to Generative Modeling	03/2024
[T11]	Invited talk for Research laboratory performance presentation, University of Seoul Generative Modeling through the Semi-dual Formulation of Unbalanced Optimal Transport	02/2024
[T12]	Seoul AI Hub Conference, AI Seoul Analyzing Optimal Transport-based Adversarial Algorithms	02/2024
[T13]	2024 Winter Workshop on PDE and Applied Mathematics, KAIST Bridging Two Distributions through Optimal Transport	01/2024
[T14]	Medical Imaging AI Leading Innovation Center, SNU Generative Modeling through the Semi-dual Formulation of Unbalanced Optimal Transport	12/2023
[T15]	Samsung Electronics Science Project Workshop, Samsung Electronics Distributional Matching between Wafer Datasets	08/2023
[T16]	Samsung Electronics Science Project Workshop, Samsung Electronics Feature Extraction from Wafer Map Datasets	03/2023

TEACHING & ASSISTANCE EXPERIENCE

Teaching Assistant

Calculus I, II, SNU	Spring 2018, Fall 2018, Fall 2019
Mathematics for Life Sciences I, II, SNU	Fall 2020, Fall 2022, Spring 2023

Course Assistant

Optimal Transport, Georgia Tech	Spring 2025
Engineering Mathematics, SNU	Spring 2020
Mathematical Analysis, SNU	Fall 2021, Spring 2022
Computer Application for Scientific Computation, SNU	Spring 2021

ACADEMIC SERVICES

Reviewer: NeurIPS (2024–present), ICLR/ICML (2025–present), CVPR (2026), Automatica (2024), TMLR (2024–2025)

CERTIFICATES

Secondary (Middle/High) School Teacher Certificate	02/2018
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